

Active Learning Strategies for Label Propagation on Graphs

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1 Introduction

In many application domains, obtaining labeled data can be resource-intensive, time consuming, and may require expert knowledge. Protein crystallography, for example, can cost upwards of \$270,000 to determine the shape of a single protein. Cybersecurity threat classification demands analysts who can distinguish between benign and malicious network activity. Document classification at scale requires subject-matter experts familiar with the taxonomy being applied. In these domains, only a small subset of data points can feasibly be labeled within practical time and cost constraints. This motivates the development of machine learning methods that can leverage a small set of labeled data points to infer labels for the remaining unlabeled data, greatly reducing the resources needed.

One approach to propagating labels across a dataset with few labeled data points is graph-based semi-supervised learning. By framing the data as a graph where data points are vertices, we can encode the similarity between data points as weighted edges. Under the smoothness assumption – that nearby or strongly connected nodes in the graph are likely to share the same label – labels assigned to a small subset of nodes can be propagated to the remaining unlabeled nodes through the graph structure.

A foundational approach in this setting is the Gaussian Fields and Harmonic Functions (GFHF) method, introduced by Zhu, Ghahramani, and Lafferty (2003). GFHF models the label assignment as a real-valued function over the graph and computes the unique harmonic function that minimizes a quadratic energy functional,

$$E(f) = \frac{1}{2} \sum_{i,j} w_{ij} (f(i) - f(j))^2,$$

subject to the constraint that f agrees with the known labels on the labeled nodes. Basically the resulting function is smooth over the graph, meaning connected nodes with high edge weight tend to receive similar predicted values. The bigger the edge weight between two nodes and the bigger the difference in their values, the higher the penalty.

While GFHF is effective at propagating labels, it does not address which nodes should be labeled in the first place. The choice of which nodes to label can have a significant impact on classification accuracy. Labeling a node that sits at the boundary between two clusters may improve predictions across a large portion of the graph, while labeling a node deep inside an already well-classified cluster would yield negligible benefit. This motivates the use of active learning strategies, which aim to select the most informative nodes to query for labels under a fixed labeling budget B . This project investigates how different query strategies affect the accuracy of GFHF-based label propagation and how robust these strategies are when labels may be incorrect.

2 Background and Literature

Zhu, Ghahramani, and Lafferty (2003) introduced the GFHF method, establishing a foundational technique in graph-based semi-supervised learning. The approach begins by constructing a graph where every data point is a node, and edges connect similar nodes. Edge weights are computed using the Gaussian kernel: the closer two points are in feature space, the higher their edge weight, and the weight drops off exponentially as distance increases. The σ parameter controls how fast this decay happens.

The central idea of GFHF is that each unlabeled node’s predicted value should equal the weighted average of its neighbors’ values. If most of a node’s neighbors lean toward class 1, that node is likely class 1 as well.

The algorithm proceeds as follows:

1. **Weighted Graph.** Begin with a weighted graph $G = (V, E)$ where each of the n data points corresponds to a node. Edge weights w_{ij} encode similarity.
2. **Gaussian Kernel.** Edge weights are computed using:

$$w_{ij} = \exp\left(-\frac{\|x_i - x_j\|^2}{2\sigma^2}\right).$$

The first l nodes are labeled with known values $f_l = (y_1, \dots, y_l)$, and the remaining $u = n - l$ nodes are unlabeled.

3. **Harmonic Function.** GFHF computes a harmonic function f over the graph that satisfies the averaging property: for each unlabeled node j ,

$$f(j) = \frac{1}{d_j} \sum_{i \sim j} w_{ij} f(i).$$

This leads to a closed-form solution:

$$f_u = (D_{uu} - W_{uu})^{-1} W_{ul} f_l.$$

4. **Energy Minimization.** The harmonic function can also be understood as minimizing the graph smoothness energy:

$$E(f) = \frac{1}{2} \sum_{i,j} w_{ij} (f(i) - f(j))^2,$$

subject to labeled nodes remaining fixed.

5. **Where:**

- D_{uu} is the degree matrix restricted to unlabeled nodes
- W_{uu} is the weight submatrix among unlabeled nodes
- W_{ul} connects unlabeled nodes to labeled nodes

Bilgic et al. (2010) developed ALFNET, an algorithm for active learning on networked data. ALFNET groups nodes into clusters based on the graph structure, then trains two classifiers – one that only uses node features and one that also considers neighbor labels. Where these classifiers *disagree* the most, that’s where a new label would help the most. They tested on Cora and CiteSeer citation networks and showed ALFNET outperforms random sampling and uncertainty sampling, proving that actively selecting nodes to label improves classification accuracy.

Fang et al. (2014) extend active learning to crowdsourcing settings where labels may be incorrect. They model multiple labelers with varying expertise and use sparse coding to learn abstract representations from unlabeled data in related domains. Their approach selects both which data point to label and which labeler to ask, choosing the most uncertain instance and the most competent labeler. While our project uses a simpler noise model, their work motivates studying how label noise interacts with query strategies.

3 Methodology

The primary goal of this project is to evaluate how different active learning strategies affect GFHF-based label propagation under limited budgets and noisy conditions.

3.1 Problem Setup

Given a weighted undirected graph $G = (V, E, W)$ with n vertices, the weight matrix W encodes pairwise similarity. The vertices are partitioned into a labeled set L and unlabeled set U , with $|L| = l \ll n$. Each labeled node has a known label $y_i \in \{0, 1\}$.

The weight matrix is constructed using the Gaussian kernel, then sparsified using k -nearest-neighbors. Label propagation uses the GFHF closed-form solution:

$$f_u = (D_{uu} - W_{uu})^{-1} W_{ul} f_l.$$

Classification is performed by thresholding: $\hat{y}_i = 1$ if $f(i) \geq 0.5$, else $\hat{y}_i = 0$.

A labeling budget B constrains the number of additional nodes that can be queried. At each iteration, a query strategy selects an unlabeled node, its label is revealed (possibly with noise), and the harmonic function is recomputed.

3.2 Active Learning Strategies

Four query strategies are implemented:

1. **Random Selection.** Baseline – select a random unlabeled node each iteration.
2. **Uncertainty Sampling.** Select the node with highest predictive entropy:

$$v^* = \arg \max_{v \in U} H(f(v)),$$

where $H(p) = -p \log p - (1 - p) \log(1 - p)$. This targets nodes near the decision boundary where the model is most unsure.

3. **Maximum Degree.** Select the unlabeled node with the highest weighted degree $d_v = \sum_j w_{vj}$. High-degree nodes influence many neighbors during propagation.
4. **Betweenness Centrality.** Select the node with highest betweenness centrality – nodes that act as bridges between different regions of the graph.

3.3 Noise Model

A symmetric label-flipping noise model is used. When querying node v , the observed label is:

$$\tilde{y}_v = \begin{cases} y_v & \text{with probability } 1 - p, \\ 1 - y_v & \text{with probability } p. \end{cases}$$

Noise rates of $p \in \{0.0, 0.05, 0.10, 0.20\}$ are tested.

4 Results

All experiments use 200 data points, $k = 10$ nearest neighbors, and a budget of $B = 20$ queries, averaged over 20 trials.

4.1 Strategy Comparison

Table 1 summarizes the final accuracies.

Dataset	Random	Uncertainty	Max Degree
Two Moons	96.18%	99.61%	93.74%
Concentric Circles	95.87%	100.00%	90.37%
Gaussian Blobs	100.00%	100.00%	100.00%

Table 1: Mean accuracy after 20 queries, averaged over 20 trials.

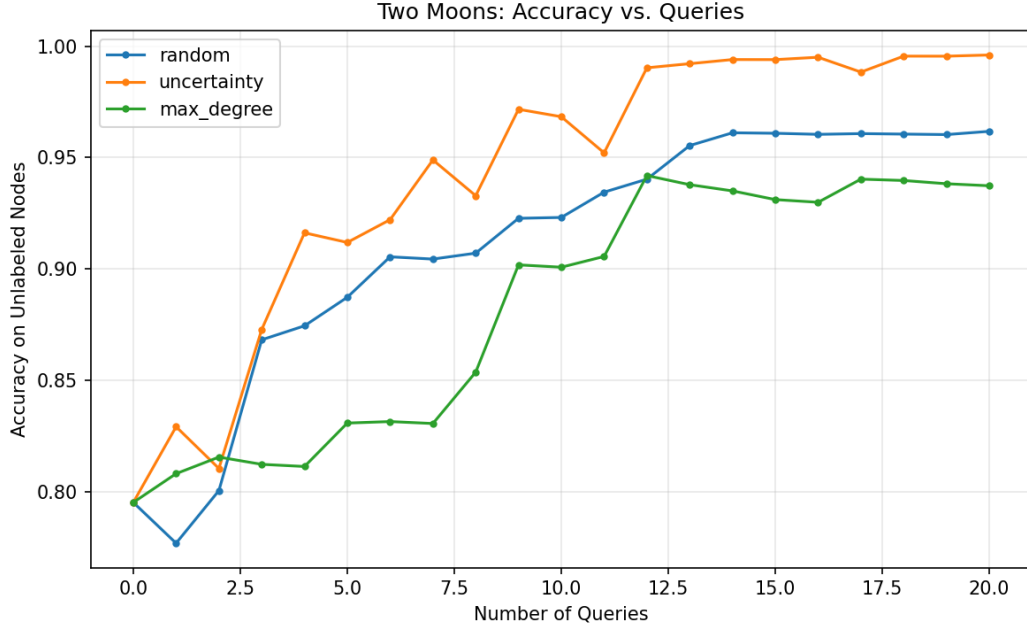


Figure 1: Two Moons: accuracy vs. queries. Uncertainty sampling converges fastest.

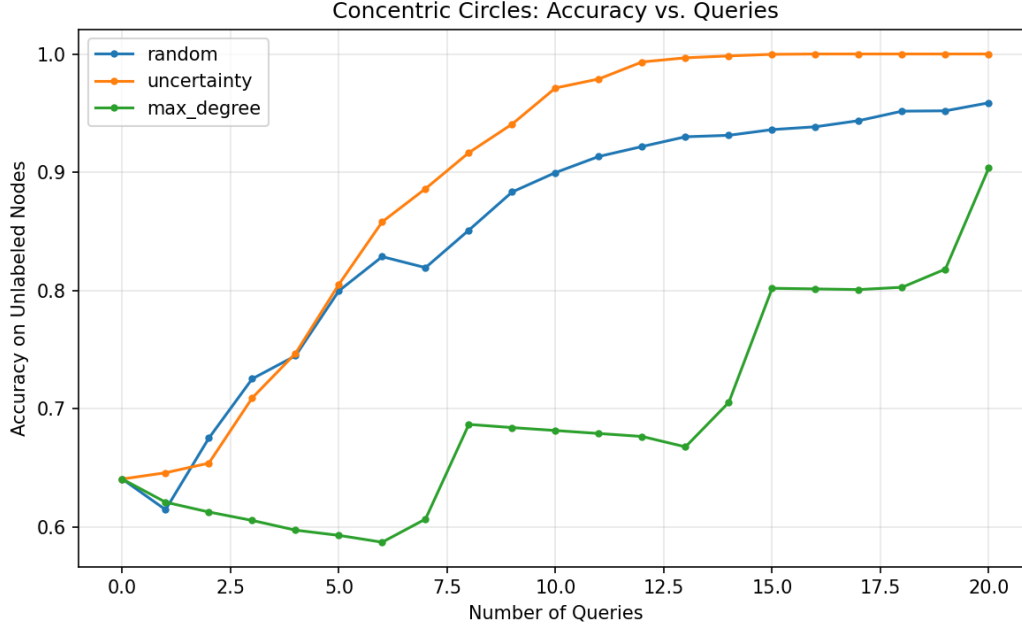


Figure 2: Concentric Circles: uncertainty sampling reaches 100% by query 12. Max degree actually decreases in accuracy during early iterations.

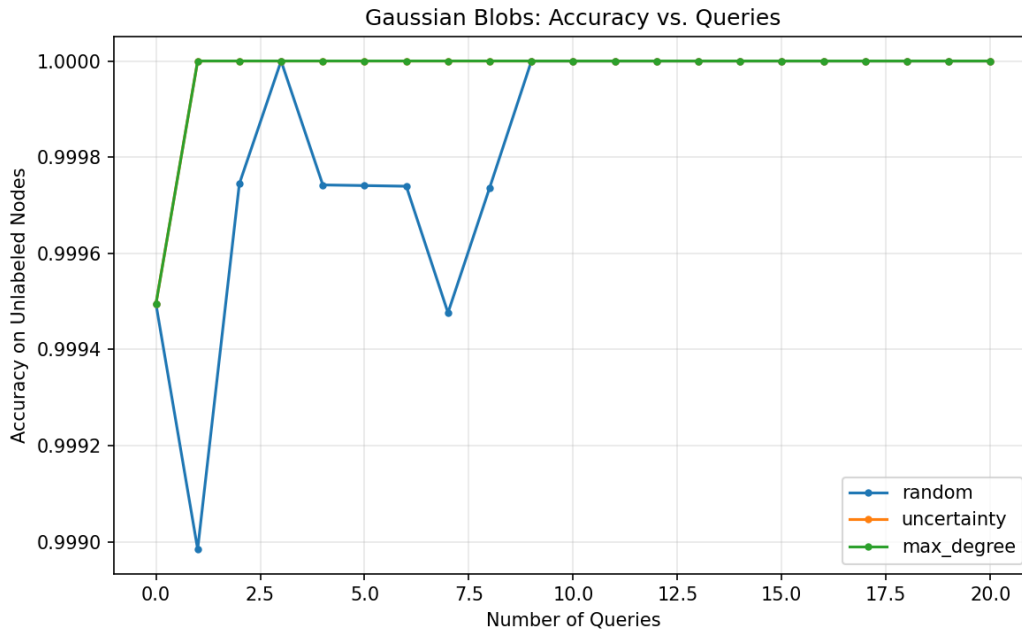


Figure 3: Gaussian Blobs: all strategies achieve near-perfect accuracy immediatly.

Uncertainty sampling outperforms the other strategies on two moons and concentric circles. On two moons it reaches 99.61% accuracy, compared to 96.18% for random and 93.74% for max degree. On concentric circles the gap is even larger – uncertainty reaches perfect classification while max degree struggles, ending at just 90.37%.

Max degree performs worse than random on both non-trivial datasets. This is because high-degree nodes tend to be deep inside clusters, not at the decision boundary. Labeling them provides redundant information.

On Gaussian blobs all strategies reach 100% because the clusters are well-separated and GFHF classifies them correctly with just the initial two labels.

4.2 Impact of Noise

Noise Rate	Random	Uncertainty	Max Degree
0%	96.18%	99.61%	93.74%
5%	93.65%	99.04%	93.12%
10%	90.76%	95.90%	90.28%
20%	86.32%	91.63%	83.90%

Table 2: Accuracy under varying noise rates on Two Moons.

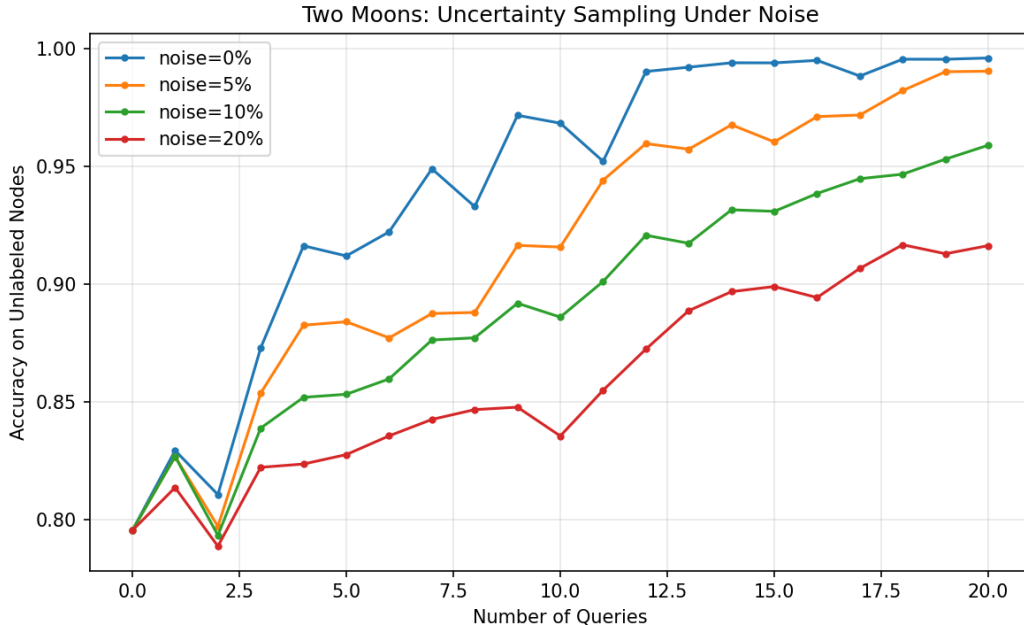


Figure 4: Uncertainty sampling under increasing noise on Two Moons.

Uncertainty sampling maintains its advantage at every noise level. Even at 20% noise it achieves 91.63% compared to 86.32% for random. The degradation is gradual – uncertainty loses about 8 percentage points going from 0% to 20% noise, while random loses about 10.

5 Conclusion

This project evaluated how active learning strategies affect GFHF label propagation. Uncertainty sampling consistently outperformed random selection and max degree selection across all datasets and noise conditions.

The key finding is that the most connected nodes are not the most useful to label. High-degree nodes sit in dense cluster interiors where GFHF already propagates labels effectively. The most valuable nodes are those at the decision boundary, where the harmonic function is most uncertain.

Uncertainty sampling also degrades gracefully under noise, losing less accuracy than random selection as noise increases. This is encouraging for real-world applications where label quality cannot be guaranteed.

Future work could implement Class Mass Normalization for imbalanced datasets, more sophisticated noise models like Fang et al.’s multi-labeler framework, and combine uncertainty with structural information as in ALFNET. Testing on real-world graph datasets like citation networks would also be valuable.

References

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